Chapter Two (Reflection and Refraction)

1-2 Reflection and refraction at a plane surface

The direction of light rays changes when they reflect from a surface, travel from one transparent medium into another or travel through a medium whose composition is continuously changing. When light travels through the different medium two phenomena happen. They are reflection and refraction. Let us now discuss each one.

1-1-2 Reflection

Reflection *is the bouncing back of the light rays from a surface in the same medium*. In other words when a ray of light comes back into the same medium after striking the surface of another medium then the phenomenon is called reflection of light. There are two types of reflection as shown in figure (1):

a) Reflection off of a smooth surface is called **specular reflection** (which we will assume from this point forward).
b) Reflection off of a rough surface is called **diffuse reflection**.

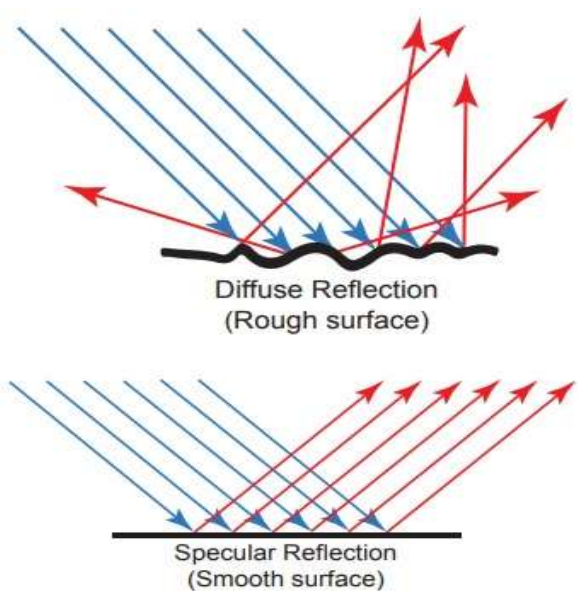


Figure (1): Types of Reflection

When light is incident on a plane mirror, most of it gets reflected, and some of it gets absorbed in the medium.

In order to understand the reflection we need to know the following:

Normal: A line drawn perpendicular to a plane at the point (P) where the ray strikes the surface.

Angle of incidence: The angle between the incident ray and the normal drawn to the point of incidence.

Angle of reflection: The angle between the reflected ray and the normal drawn to the point of incidence.

❖ **<u>The laws of reflection states that:</u>**

1. The angle of incidence (θ_i) is identical to the angle of reflection (θ_r). (See figure (2))

   $\theta_i = \theta_r$ ……..…….(1)

   The angles are measured relative to Normal.

2. The ray of incidence, the ray of reflection, and the normal to the mirror, occurs on the same plane. (See figure (2)).

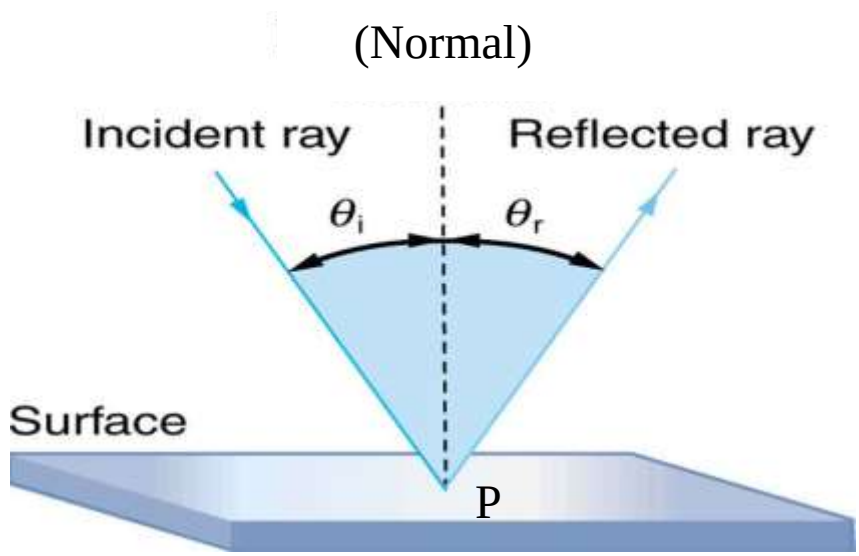


Figure (2): Reflection of Light ray

Prove of Reflection Law using Huygen's principle

Plane AB is incident wavefront at an angle (θ_i) on the reflecting surface MN. Let the ray B reach the surface MN at point C after time τ as shown in figure (3). If v is the speed of light in air then:

BC= $v\tau$ ……. (a)

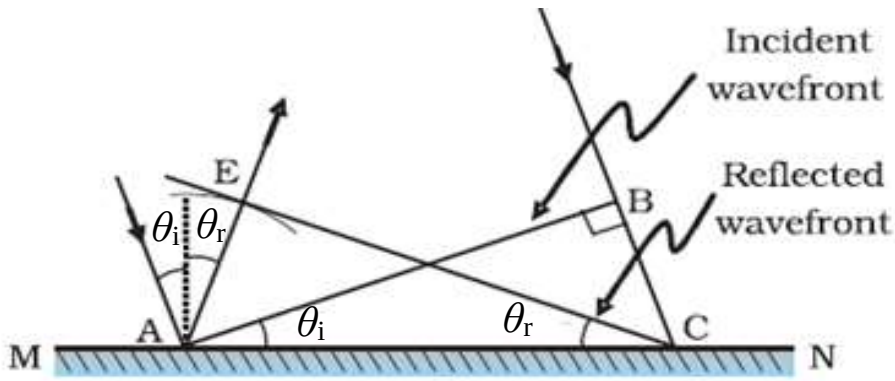


Figure (3): Reflection of Light by Huygens

During the time τ the secondary wavelet from point A spreads over a hemisphere of radius $v\tau$ with center at A. let CE be the tangent to this hemisphere. Then:

AE=BC …….…. (b)

Also C and E are in the same phase. Hence CE is the reflected wavefront.

Draw normal to MN at the point A, then

θ_i is the angle of incidence, and θ_r is the angle of reflection

In triangles Δ ABC and Δ AEC we have:

AE=BC; AC is common,

and <ABC= <AEC=90° ……... (c) (rays are normal to wavefronts)

∴Δ ABC and Δ AEC are congruent

∴ $\theta_i = \theta_r$ Which is Reflection Law

2-1-2 Refraction

The change of direction and speed of light as it enters from one medium to another is termed as the **Refraction** of light. Examples of refraction are lenses, prism, and telescope. Refraction means that the light beam bends. This bending takes place because the light beam's (i.e., photon's) velocity changes as it goes from one medium to the next.

Angle of refraction: The angle between the refracted ray and the normal drawn to the point of incidence

Rule -1: When a light ray travels from a rarer medium to a denser medium, the light ray bends towards the normal as velocity of light decreases with respect to rare medium.

Rule -2: When a light ray travels from a denser medium to a rarer medium, the light ray bends away from the normal as velocity of light increases with respect to denser medium. As shown in figure (4) and (5).

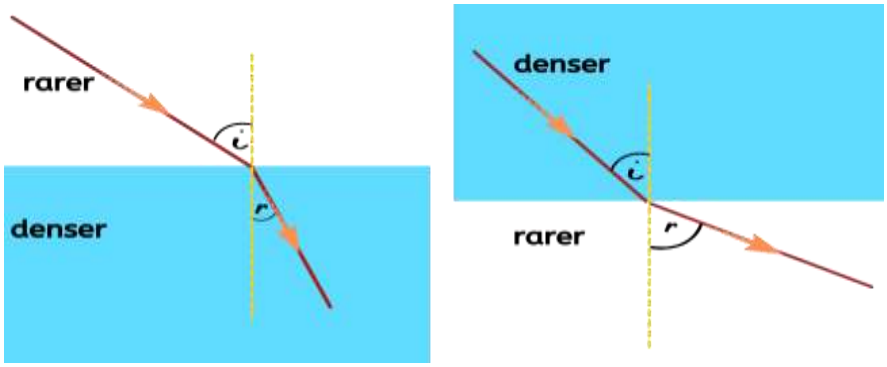


Figure (4): Refraction of Light between rarer and denser

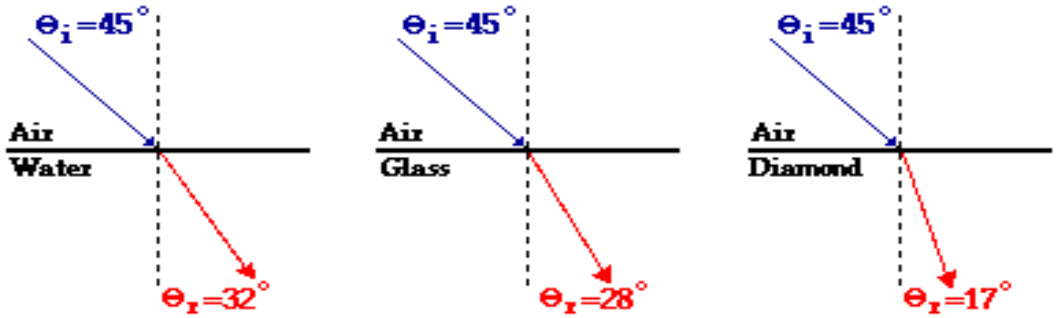


Figure (5): Reflection of Light between different mediums

❖ **<u>The laws of refraction states that:</u>**

1- The incident ray, the refracted ray and normal ray at the interface of any two given medium, all lie in the same plane.

2- The ratio of the sine of the angle of incidence (θ_i) and the sine of the angle of refraction (θ_r) is constant in any two given medium.

$$\frac{sin\theta_r}{sin\theta_i} = \frac{v_r}{v_i} = constant \text{ .....(2)}$$

Where:

v_r and θ_r are the velocity and the angle of the refracted beam with respect to the normal line of the surface.

v_i and θ_i are the velocity and the angle of the incident beam with respect to the normal line of the surface.

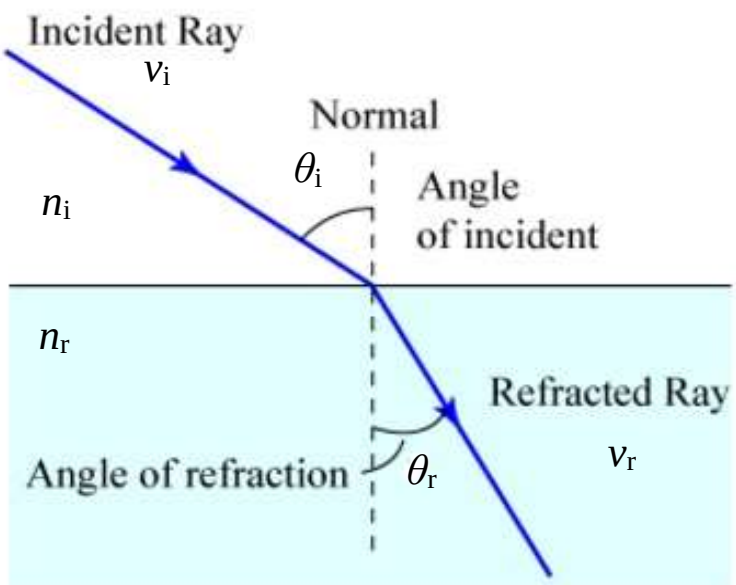


Figure (3): Refraction of Light ray

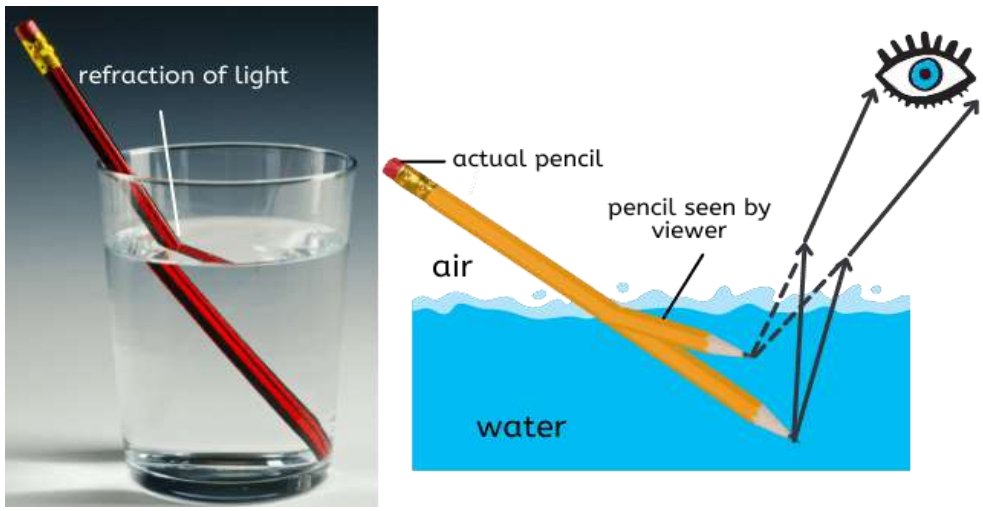


Figure (4): Example of Refraction of Light

As we mentioned in chapter one the law of refraction of index is:

$$n = \frac{\text{speed of light in vacuum}}{\text{speed of light in the medium}} = \frac{c}{v}$$

Applying this equation for the incident medium we get:

$$n_i = \frac{c}{v_i} \longrightarrow v_i = \frac{c}{n_i} \ldots\ldots\ldots\ldots(3)$$

And for the refracted medium we get:

$$n_r = \frac{c}{v_r} \longrightarrow v_r = \frac{c}{n_r} \ldots\ldots\ldots\ldots(4)$$

Putting eqs. (3) and (4) in eq. (2) we have:

$$\frac{sin\theta_r}{sin\theta_i} = \frac{\frac{c}{n_r}}{\frac{c}{n_i}} = \frac{n_i}{n_r}$$

$$\boxed{n_i sin\theta_i = n_r sin\theta_r} \ldots\ldots\ldots\ldots(5) \quad \{\text{ Snell's Law }\}$$

Prove of Refraction Law (Snell's Law) using Huygen's principle

Let PP' represent a surface separating the medium 1 and medium 2. The speed of the light in these mediums are v_1 and v_2 respectively. A ray of light AA' is incident on this surface and another ray which is parallel to this ray is also incident on this surface. If we draw a perpendicular from point (A) to the another ray of light, Point A, and point B will have a line joining them and this is called as wavefront which is incident on the surface PP' as shown in figure (5)

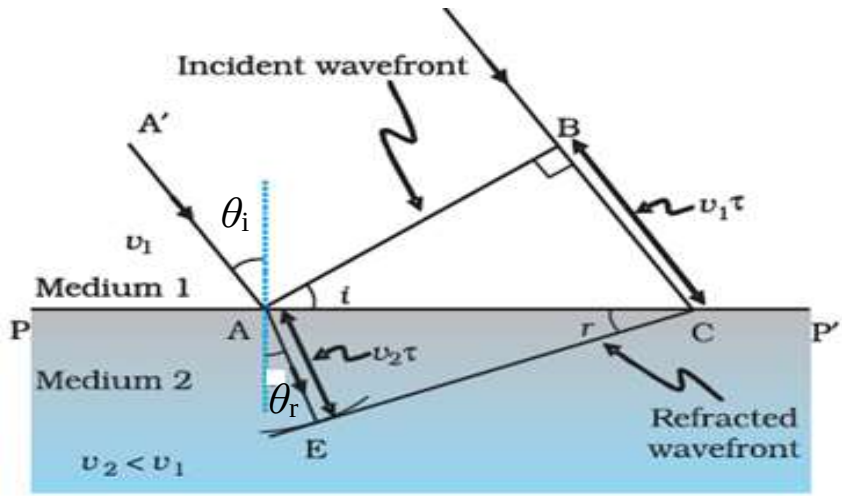


Figure (5): Refraction of Light by Huygens

Since Huygen's Principle states that despite differences in density, the time taken by the waves to travel will be the same, let's assume the time taken is τ.

Therefore, distance:
BC = $v_1\tau$ and AE = $v_2\tau$ ……..(a)
and CE would represent the refracted wavefront.
Considering the triangles Δ ABC and Δ AEC, we get:

$sin\theta_i = \frac{BC}{AC} = \frac{v_1\tau}{AC}$ …… (b)

$sin\theta_r = \frac{AE}{AC} = \frac{v_2\tau}{AC}$ …… (c)

$\frac{sin\theta_i}{sin\theta_r} = \frac{BC}{AE} = \frac{v_1}{v_2} = \frac{n_2}{n_1}$ ……... (d)

$n_1 sin\theta_i = n_2 sin\theta_r$ ……..… (e)

Which is Snell's Law

Example (1): Light travels from air into an optical fiber with an index of refraction of 1.44

(a) In which direction does the light bend?

(b) If the angle of incidence on the end of the fiber is 22°, what is the angle of refraction inside the fiber?

(c) Sketch the path of light as it changes media.

Solution:

a) Since the light is traveling from a rarer region (lower n) to a denser region (higher n), it will bend toward the normal.

b) We will identify air as medium (1) and the fiber as medium (2).

Thus, $n_1 = 1.00$, $n_2 = 1.44$, and $\theta_1 = 22^o$.

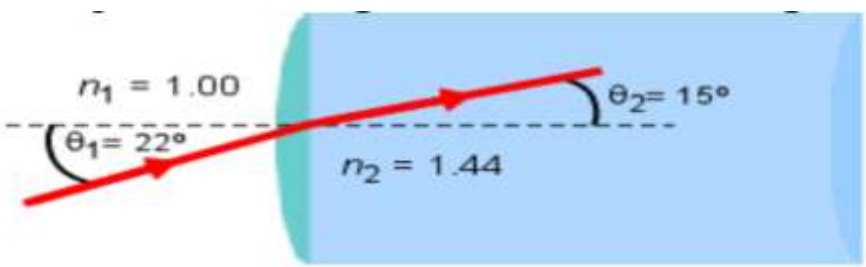
Snell's Law then becomes:

$(1.00) \sin 22^o = 1.44 \sin \theta_2$

$\sin \theta_2 = (1.00/1.44) \sin 22^o = 0.260$

$\theta_2 = \sin^{-1} (0.260) = 15^o$

c) The path of the light is shown in the figure below



Example (2): Light traveling through an optical fiber (n=1.44) reaches the end of the fiber and exits into air.

(a) If the angle of incidence on the end of the fiber is 30°, what is the angle of refraction outside the fiber?

(b) How would your answer be different if the angle of incidence were 50°?

Solution:

(a) Since the light is now traveling from the fiber into air, we will call the fiber material (1) and air material (2).

Thus, $n_1 = 1.44$, $n_2 = 1.00$, and $\theta_1 = 30^o$.

Snell's Law then becomes:

(1.44) sin 30° = 1.00 sin θ_2
sin θ_2 = (1.44/1.00) sin 30° = 1.44 (0.50) = 0.720
$\theta_2 = \sin^{-1}(0.720) = 46°$
Notice that this time, the angle of refraction is larger than the angle of incidence. The light is bending away from the normal as it enters a rarer material.

(b) Replacing the angle of incidence with 50° gives

sin θ_2 = (1.44/1.00) sin 50° = 1.44 (0.766) = 1.103

This equality cannot be met, so light cannot exit the fiber under these conditions.

Example (3): A ray of light strikes a flat, 2.00 cm thick block of glass (n_g = 1.50) at an angle of 30° with respect to the normal (as drawn in the figure below).

(a) Find the angle of refraction at the top surface.

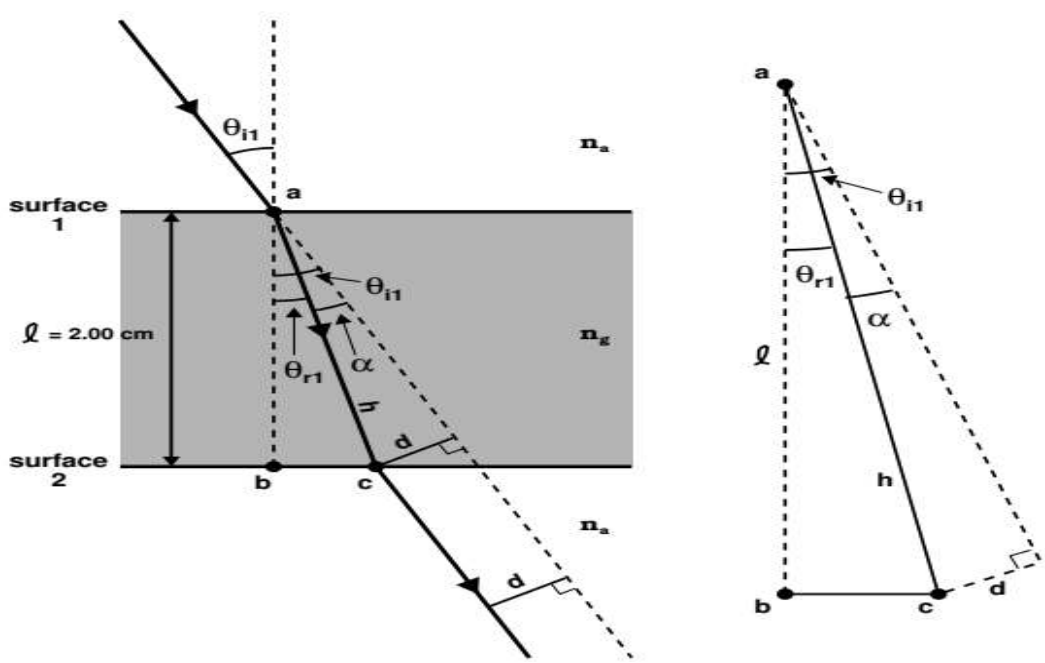
(b) Find the angle of incidence at the bottom surface and the refracted angle at this surface.

(c) Find the lateral distance d by which the light beam is shifted.

(d) Calculate the speed of light in the glass and

(e) Find the time required for the light to pass through the glass block.

(f) Is the travel time through the block affected by the angle of incidence? Explain.



Solution:

(a) At the top surface (labeled 1), the angle of incidence is given to us as $\theta_{i1} = 30^o$. We will assume that the beam is hitting the glass from air, hence $n_a = 1.00$. Snell's law gives the refracted angle in the glass (with $n_g = 1.50$) at surface (1) as:

$$n_a \sin \theta_{i1} = n_g \sin \theta_{r1}$$

$$sin\theta_{r1} = \frac{n_a sin\theta_{i1}}{n_g}$$

$$\theta_{r1} = sin^{-1}\left[\frac{n_a sin\theta_{i1}}{n_g}\right]$$

$$= sin^{-1}\left[\frac{(1.00)sin30^o}{1.50}\right] = 19.5^o$$

(b) Since the second surface (labeled 2) is parallel to the first, the angle of incidence at the bottom surface is exactly the same as the refracted angle at the top surface following the theorem of geometry, so $\theta_{i2} = \theta_{r1} = 19.5^o$. The angle of refraction at this bottom surface (back into the air) is then:

$$n_g \sin \theta_{i2} = n_a \sin \theta_{r2}$$

$$sin\theta_{r2} = \frac{n_g sin\theta_{i2}}{n_a}$$

$$\theta_{r2} = sin^{-1}\left[\frac{n_g sin\theta_{i2}}{n_a}\right]$$

$$= sin^{-1}\left[\frac{(1.50)sin19.5^o}{1.00}\right] = 30.0^o$$

Thus, the light emerges traveling parallel to the incident beam.

(c) Let ℓ 2.00 cm be the thickness of the glass. The angle of refraction at the first surface from Part (a) is $\theta_{r1} = 19.5^o$. Let (h) represent the distance from point (a) to (c) (i.e., the hypotenuse of triangle (abc), then:

$$h = \frac{\ell}{cos\theta_{r1}} = \frac{2.00\ cm}{cos19.5} = 2.12\ cm$$

From the figure the angle $\alpha = \theta_{i1} - \theta_{r1} = 30.0^o - 19.5^o = 10.5^o$

Then: $d = h \sin \alpha = (2.12 \text{ cm}) \sin 10.5^o = 0.386 \text{ cm}$

(d) The speed of light in the glass is given by:

$$v = \frac{c}{n_g} = \frac{3\times10^8 m/s}{1.50} = 2 \times 10^8\, m/s$$

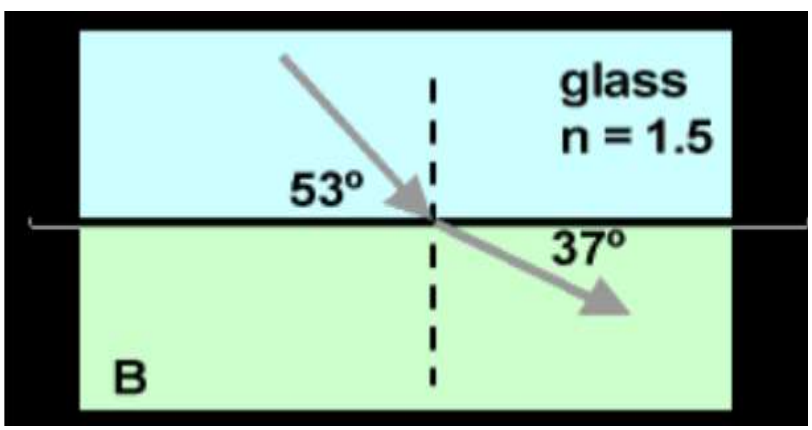
(e) The time it will take the light beam to travel through the glass is therefore:

$$t = \frac{h}{v} = \frac{2.12\times10^{-2} m}{2\times10^8 m/s} = 1.06 \times 10^{-10} s = 0.106 ns$$

(f) If the angle of incidence θ_{i1} on the top surface is changed, the angle of refraction for that surface θ_{r1} will change and the path length (h) that the light beam takes will be modified base on the equation for h on the previous page. Since the value for h changes the time it takes for the light beam to travel through the glass will change based upon the equation above.

Home Work:

Calculate the index of refraction for medium B.



2-2 Optical path length (OPL)

The path (geometric length) *d* of a ray of light in any medium is calculated by taking the product of *velocity* and *time:*

$d = vt \longrightarrow v = \frac{d}{t}$ ……..(6)

From the definition of refraction index we have:

$n = \frac{c}{v} \longrightarrow v = \frac{c}{n}$

Then :

$\frac{d}{t} = \frac{c}{n}$

So we can write:

$nd = ct$ …….……..(7)

The product (*nd*) is called the **optical path length (OPL)** Δ:

$\Delta = nd$ ……..……(8)

The optical path represents the distance light travels in a vacuum in the same time it travels a distance *d* in the medium. If a light ray travels through a series of optical media of thickness d_1, d_2, and d_3,.... and refractive indices n_1, n_2, and n_3,..., the total optical path is just the sum of the separate values:

$\Delta = n_1 d_1 + n_2 d_2 + n_3 d_3 + \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots$ (9)

Figure (6) shows the meaning of optical path.

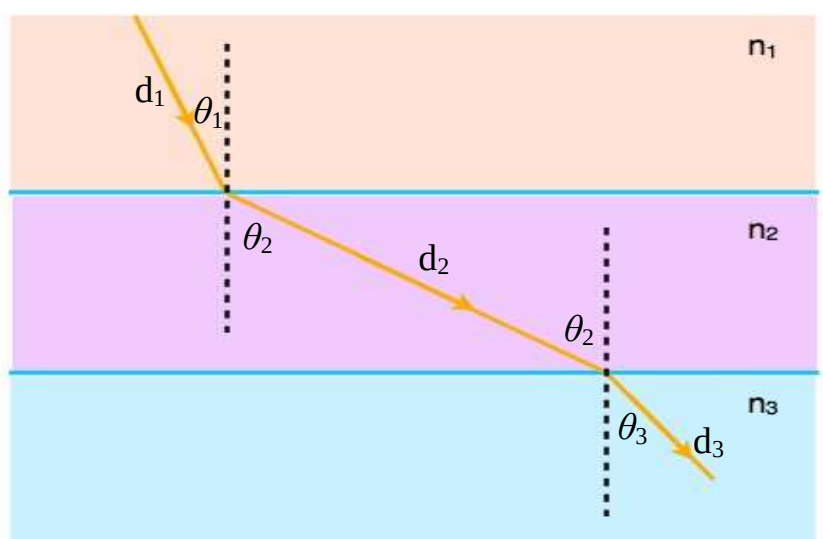


Figure (6): Optical path length for series of mediums

A difference in OPL between two paths is often called the **optical path difference** (OPD).

3-2 Fermat's principle:

States that *light travels between two points along the path* (*Optical path length*) *that requires the least time, as compared to other nearby paths*. A useful results of Fermat's Principle that it's proved that in any medium where the index of refraction is constant, light travels in a straight line. Also prove the Law of Reflection.

1-3-2 Derivation of reflection law according to Fermat's principle:

Consider the light ray as shown in figure (7), a ray of light starting from point A .We calculate the length of each path and divide them by the speed of light to determine the time required to travel between two points. The path length from A to B is

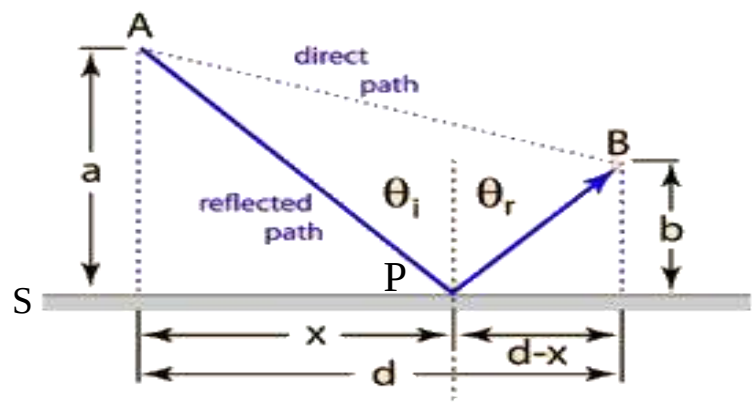


Figure (7): The path of reflected light ray

The path length from A to B is:

L= AP+PB ……..…(10)

Using **Pythagorean Theorem** and dividing by the speed of light we have the time required to travel between two points:

$$t = \frac{L}{c} = \frac{\sqrt{a^2+x^2}}{c} + \frac{\sqrt{b^2+(d-x)^2}}{c} \quad \ldots\ldots\ldots\ldots\ldots(11)$$

To minimize the time we set the derivative of the time with respect to x equal to zero (Fermat's principle).

$$\frac{dt}{dx} = \frac{1}{2}\frac{2x}{c\sqrt{a^2+x^2}} + \frac{1}{2}\frac{2(d-x)(-1)}{c\sqrt{b^2+(d-x)^2}} = 0 \quad \text{.....(12)}$$

This reduces to:

$$\frac{x}{\sqrt{a^2+x^2}} = \frac{(d-x)}{\sqrt{b^2+(d-x)^2}} \quad \text{.....(13)}$$

Using the diagram in figure (7) we can write:

$$sin\theta_i = \frac{x}{\sqrt{a^2+x^2}} \quad \text{and} \quad sin\theta_r = \frac{(d-x)}{\sqrt{b^2+(d-x)^2}}$$

Then : $\sin\theta_i = \sin\theta_r$ .....(14)

For small angles we can write;

$$\boxed{\theta_i = \theta_r} \quad \text{Reflection Law}$$

2-3-2 Derivation of refraction law according to Fermat's principle:

Let the plane S be the interface between medium 1 and medium 2 with the refractive indices $n_1 = c/v_1$ and $n_2 = c/v_2$. Assume the light ray propagate from point A to point B across the boundary between the two mediums as shown in figure (8).

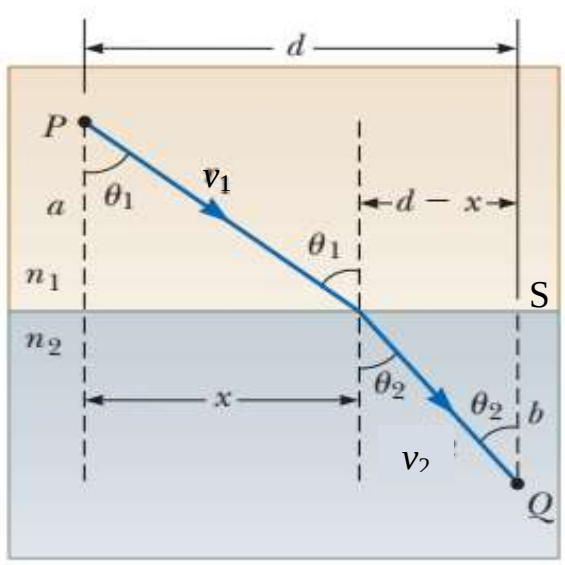


Figure (8): The path of refracted light ray

The total time needed for the light to travel from A to B is:

$$t = \frac{\sqrt{a^2+x^2}}{v_1} + \frac{\sqrt{b^2+(d-x)^2}}{v_2} \quad \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots(15)$$

Having the derivative of *t* with respect to *x* then equating it to zero we get:

$$\frac{dt}{dx} = \frac{x}{v_1\sqrt{a^2+x^2}} - \frac{(d-x)}{v_2\sqrt{b^2+(d-x)^2}} = o \quad \ldots\ldots\ldots\ldots\ldots\ldots(16)$$

From figure (8) we have:

$$sin\theta_1 = \frac{x}{\sqrt{a^2+x^2}} \quad \text{and} \quad sin\theta_2 = \frac{(d-x)}{\sqrt{b^2+(d-x)^2}}$$

Then:

$$0 = \frac{sin\theta_1}{v_1} - \frac{sin\theta_2}{v_2} \quad \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots(17)$$

So we get:

$$\frac{sin\theta_1}{c/n_1} = \frac{sin\theta_2}{c/n_2} \quad \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots(18)$$

$$\boxed{n_1 sin\theta_1 = n_2 sin\theta_2} \quad \text{Snell's Law}$$

4-2 The principle of Reversibility

Any actual ray of light in an optical system, if reversed in direction, will retrace the same path backward

Reversing the direction of a light ray does not affect its attenuation during its passage through an optical medium due to reflection, refraction, and absorption, according to the principle of reversibility.

5-2 Total internal reflection

Total internal reflection occurs only when light attempts to move from a medium of high index of refraction to a medium of lower index of refraction, so the angle of refraction is greater than the incident angle.

The critical angle (θ_c) for the boundary separating two optical media is defined as *the smallest angle of incidence, in the medium of greater index, for which light is totally reflected*. It is determined by:

$n_1 \sin \theta_c = n_2 \sin 90^o = n_2$ ……..(19)

or $\quad sin\theta_c = \frac{n_2}{n_1}$

$\therefore\ \theta_c = sin^{-1}\left(\frac{n_2}{n_1}\right)$……..…….(20)

As can be seen, this only occurs when $n_2 < n_1$, since [$\sin \theta_c \leq 1$] ,see figure (9).

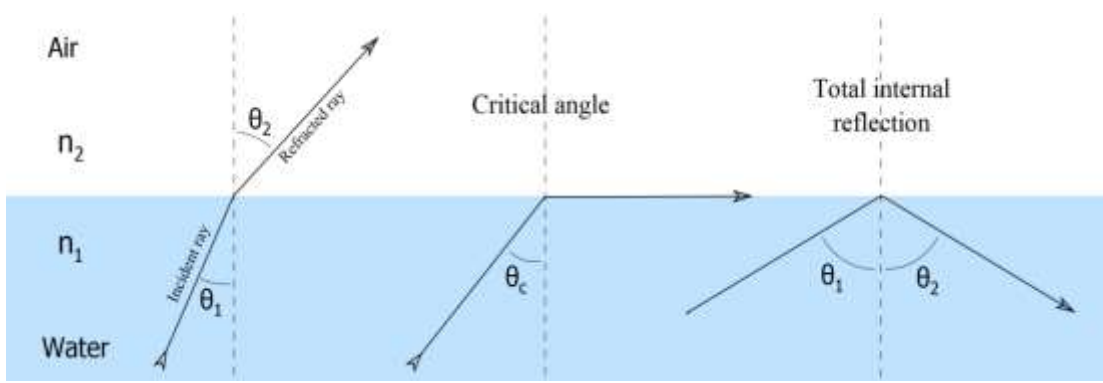


Figure (9): Total internal reflection

For incident angles greater than the critical angle, and no refracted ray exists. This phenomenon is called total internal reflection.

If the index of refraction of the glass is typical 1.5, the critical angle for total internal reflection is 41.8° where $\theta_t = 90^o$.

$$sin\theta_c = \frac{1}{1.5} sin\theta_t$$

$$\theta_c = sin^{-1}\left(\frac{1}{1.5}\right) = 41.8^o$$

If the angle of incidence of the ray on the glass–air interface is 45º, the light will be totally reflected.

Total reflection is really total in the sense that no energy is lost upon reflection.

❖ Applications of Total Internal Reflection

1- Totally Reflecting prisms: which are glass prisms with two angles of 45° and one of 90°. As shown in figure (10)

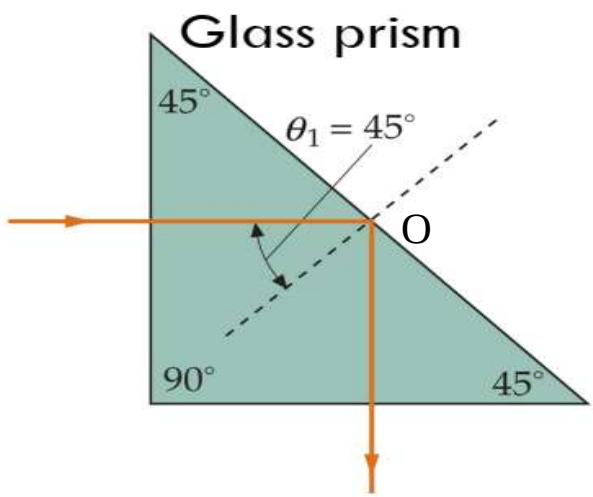


Figure (10): Total reflection prisms

The symmetry of the prism allows light to be incident on O at an angle of 45º, which is greater than the critical angle for glass i.e. 42º. As a result, light suffers total internal reflection and is deviated by 90º

2-Mirage : It is an optical illusion which is observed in deserts or on tarred roads in hot summer days. This, you might have observed, creates an illusion of water, which actually is not there. Due to excessive heat, the road gets very hot and the air in contact with it also gets heated up. The densities and the refractive indices of the layers immediately above the road are lower than those of the cooler higher layers. Since there is no abrupt change in medium (see Figure (11)), a ray of light from a distant object, such as a tree, bends more and more as it passes through these layers. And when it falls on a layer at an angle greater than the critical angle for the two consecutive layers, total internal reflection occurs. This produces an inverted image of the tree giving an illusion of reflection from a pool of water

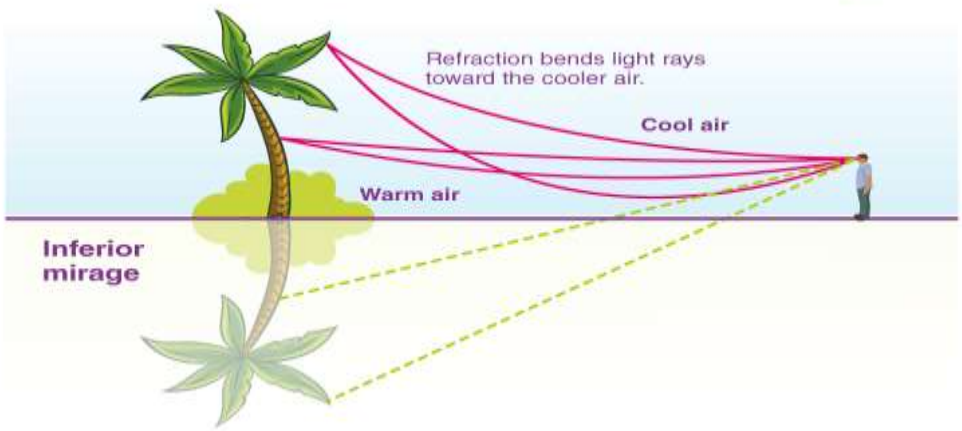


Figure (11): Mirage

3- **Fiber Optics:** is the technology used to transmit information as light pulses along a strands of fiber made of glass or plastic fiber (see figure (12)). Fiber optics is used for long-distance and high-performance data networking. It is also commonly used in telecommunication services, such as internet, television and telephones.

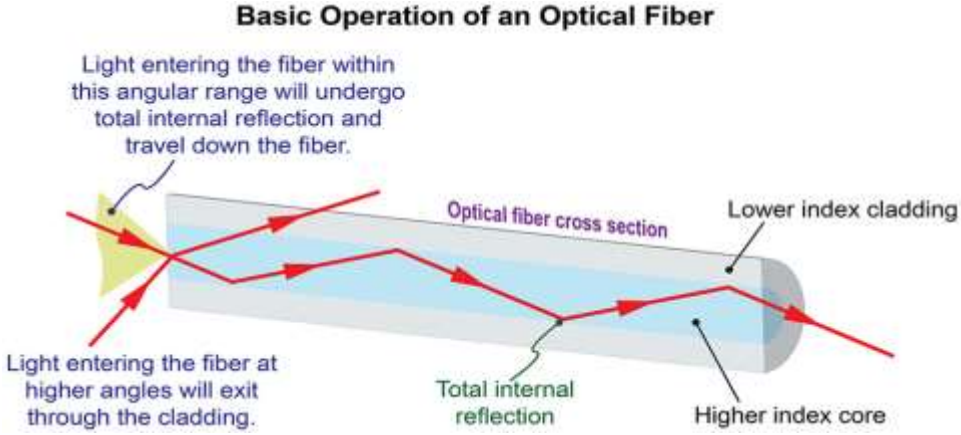
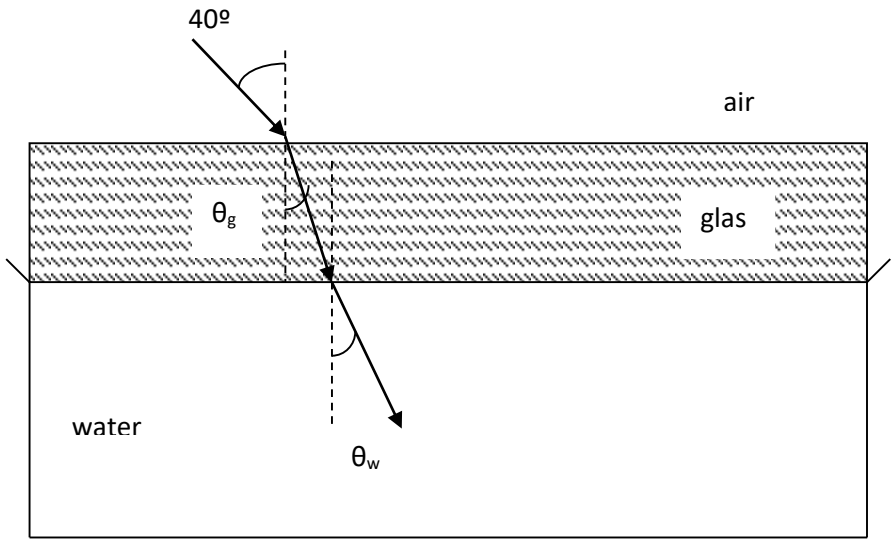


Figure (12): Fiber optics

Example (1): The speed of light in a particular piece of glass is 2.0 x 10^8 m/s, and the speed of light in water is 2.3 x 10^8 m/s.

(a) Find the index of refraction for the glass and water

(b) A sheet of this glass is placed over a tank full of water. Laser light is incident on the glass from the air above the glass at an angle of 40º. Determine whether or not the light passes into the water, and, if it does, find the angle of refraction of the light in the water.



Solution:

(a) $n_g = \frac{c}{v_g} = \frac{3\times10^8\ m/s}{2.0\times10^8\ m/s} = 1.5$

$n_w = \frac{c}{v_w} = \frac{3\times10^8\ m/s}{2.3\times10^8\ m/s} = 1.3$

(b) First, let's find the critical angle for the light traveling from the glass to the water.

$$\theta_c = sin^{-1}\left(\frac{n_w}{n_g}\right) = \sin^{-1}\left(\frac{1.3}{1.5}\right) = 60^o$$

If the light passes from the glass toward the water at an angle greater than 60º, it will totally internally reflect inside the glass. The angle of refraction inside the glass can be found by

$$n_{air} sin\theta_{air} = n_g sin\theta_g$$

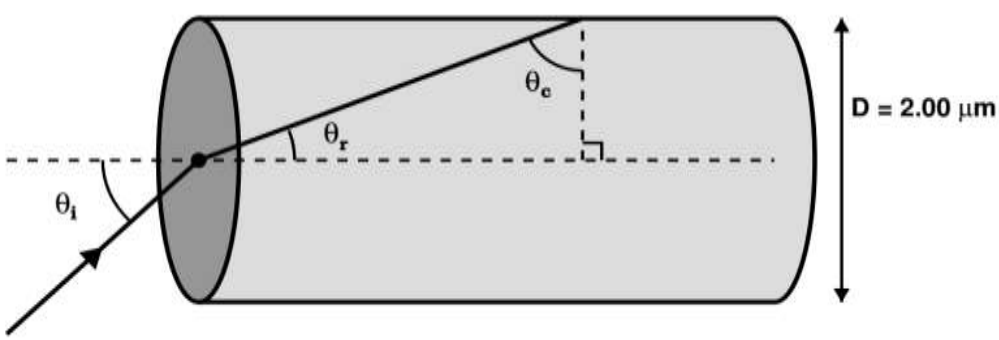
$$\theta_g = sin^{-1}\left(\frac{n_{air} sin\theta_{air}}{n_g}\right) = sin^{-1}\left(\frac{sin40}{1.5}\right) = 25.4^o$$

Since the angle of the light in the glass less than 60º, the light will refract in the water. The angle of refraction in the water can be found by

$$n_g sin\theta_g = n_w sin\theta_w$$

$$\theta_w = sin^{-1}\left(\frac{n_g sin\theta_g}{n_w}\right) = sin^{-1}\left(\frac{(1.5)\ sin(25.4)}{1.3}\right) = 29.7^o$$

Example (2): Determine the maximum angle θ_i for which the light rays incident on the end of the light pipe as shown in the figure below are subject to total internal reflection along the walls of the pipe. Assume the light pipe has an index of refraction of 1.36 and that the outside medium is air.



Note: $\theta_r + \theta_c = 90^o$

Solution:

The critical angle for this material in air is:

$$\theta_c = sin^{-1}\left(\frac{n_{air}}{n_{pipe}}\right) = sin^{-1}\left(\frac{1.00}{1.36}\right) = 47.3^o$$

Thus, the refracted angle of the light beam entering the cross sectional area of the pipe (see diagram above) is:

θ_r = 90.0º - θ_c = 42.7◦ and from Snell's law, the angle of incidence at this surface is:

$$\theta_i = sin^{-1}\left(\frac{n_{pipe} sin\theta_r}{n_{air}}\right) = sin^{-1}\left(\frac{(1.36)sin42.7^o}{1.00}\right) = 62.7^o$$

6-2 Application of Reflection and Refraction in 2-dimension

1-6-2 Plane - Parallel Plate

The laws of reflection and refraction can be used to examine the passage of light through a plate with plane parallel surfaces. Figure (13) shows an incident ray and reflected and refraction ray at the boundaries between the air and the acrylic. The light ray emerges from the plate parallel to the original ray but with a lateral displacement *d*.

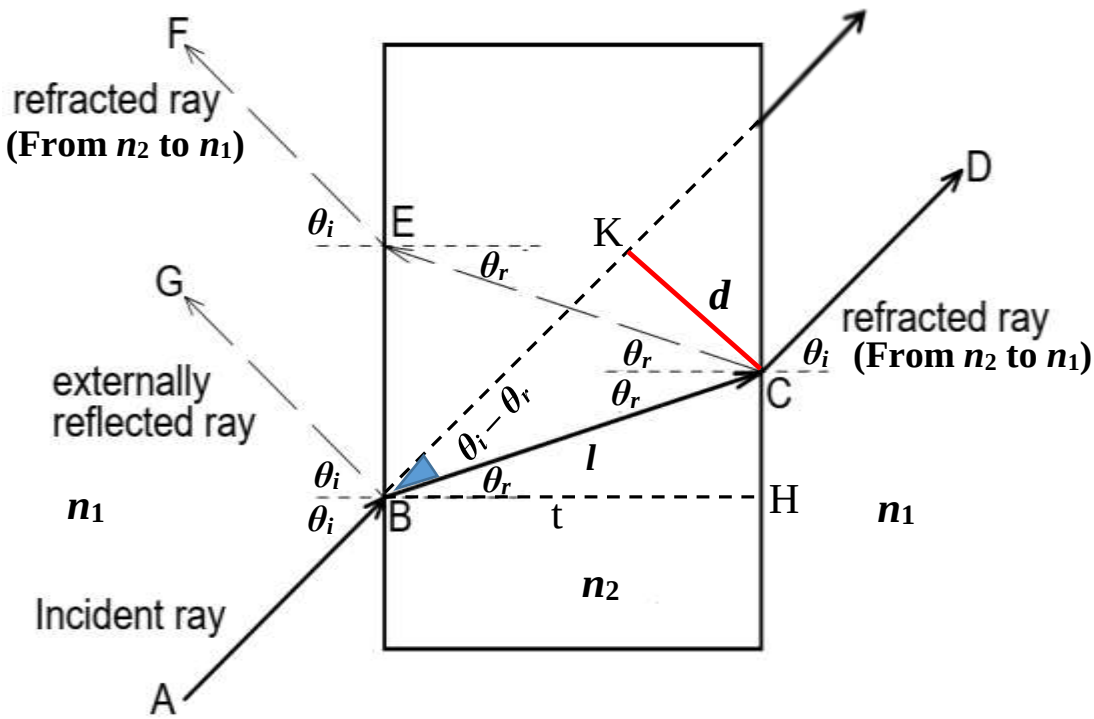


Figure (13): reflection and refraction by a plane parallel surfaces

Incident ray AB partly reflects externally to generate the reflected ray BG. The rest of the incident light refracts to ray BC. At the other end of the plate, ray BC internally reflects to give ray CE and refracts to produce ray CD, the exiting light. Ray CE behaves similarly and yields a refracted ray and an internally reflected ray which in turn gives a refracted and reflected component and so on. Because the sides of the plate are parallel, rays AB and CD are parallel, rays BG and EF are parallel, all internal angles are equal to the refraction angle θ_r , and all external angles are equal to the incident angle θ_i .

Using figure (13), we may apply the law of refraction and some simple trigonometry to find the displacement *d*.

In Δ BCK we have:

$$sin(\theta_i - \theta_r) = \frac{d}{l} \longrightarrow d = l\, sin(\theta_i - \theta_r) \;\ldots\ldots\; (21)$$

And from the right triangle Δ BHC we have:

$$\cos\theta_r = \frac{t}{l} \longrightarrow l = \frac{t}{\cos\theta_r} \;\ldots\ldots\ldots\ldots\ldots\ldots\ldots\; (22)$$

Where *t* is the thickness of the parallel plate

Substitution eq. (22) in eq. (21) we get:

$$\boxed{d = \frac{t}{\cos\theta_r}\, sin(\theta_i - \theta_r)} \;\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\; (23)$$

$$\because\; \sec\theta_r = \frac{1}{\cos\theta_r} \;\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\; (24)$$

$$\therefore \boxed{d = t\, sec\theta_r\, sin(\theta_i - \theta_r)} \;\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\; (25)$$

Lateral displacement equation (eq. (23)) can be written in another form by using the trigonometric relation for the sine of the difference between two angles as follows:

$$sin(\theta_i - \theta_r) = (sin\theta_i\, cos\theta_r - sin\theta_r\, cos\theta_i) \;\ldots\ldots\ldots\; (26)$$

Butting eq. (26) in eq. (23) we get:

$$d = \frac{t}{\cos\theta_r}(sin\theta_i\, cos\theta_r - sin\theta_r\, cos\theta_i)$$

$$d = t\left(\frac{sin\theta_i\, cos\theta_r}{\cos\theta_r} - \frac{sin\theta_r\, cos\theta_i}{\cos\theta_r}\right)$$

$$d = t\left(sin\theta_i - \frac{sin\theta_r\, cos\theta_i}{\cos\theta_r}\right) \;\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\; (27)$$

From Snell's law we obtain:

$$sin\theta_r = \frac{n_1}{n_2}\, sin\theta_i \quad \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots \text{(28)}$$

Butting eq. (28) in eq. (27) we get:

$$d = t\left(sin\theta_i - \frac{n_1 \sin\theta_i\, cos\theta_i}{n_2 \cos\theta_r}\right)$$

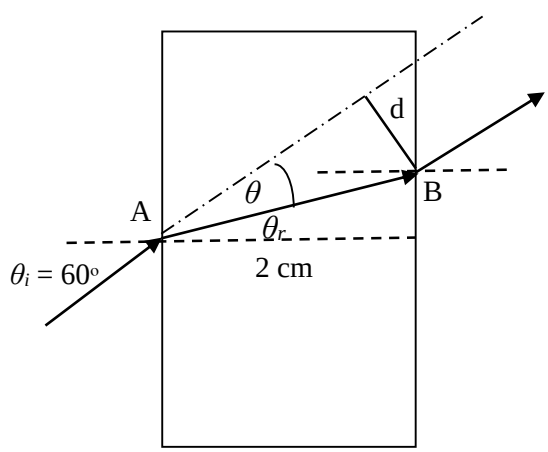
$$\boxed{d = t\, sin\theta_i \left(1 - \frac{n_1 cos\theta_i}{n_2 \cos\theta_r}\right)} \quad \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots \text{(29)}$$

Equation (29) represents the Lateral displacement equation.

<u>Example 1:</u>

Light is incidence at an angle of 60° on plane –parallel glass plate of 2 cm thickness and 1.5 refractive index surrounded by air. Find the lateral displacement between the incident and refractive ray?

Solution:



The lateral displacement (*d*) between the incident and refractive ray

$$\sin\theta = \frac{d}{AB} \longrightarrow d = AB \sin\theta$$

Where $\theta = 60 - \theta_r$

By applying Snell's law at point A to find θ_r we have:

$$n_1 sin\theta_i = n_2 sin\theta_r$$

$1 \times \sin 60 = 1.5 \sin \theta_r$

$\text{Sin } \theta_r = 0.577$

$\theta_r = \sin^{-1}(0.577) = 35.24^o$

$\because \theta = 60 - \theta_r$

$= 60 - 35.24 = 24.76^o$

In the above figure:

$$\cos\theta_r = \frac{2}{AB}$$

$$\therefore AB = \frac{2}{cos\theta_r} = \frac{2}{\cos 35.24} = \frac{2}{0.8167} = 2.49\ cm$$

Lateral displacement $d = AB \sin\theta = 2.49\ sin\ 24.76 = 1.04\ cm$

2-6-2 Refraction by a Prism

A prism is an optical element. It can be made from materials like glass, plastic and fluorite. It can be used to split light into its components.

In a prism the two surfaces are inclined at some angle so that the deviation produced by the first surface is not annulled by the second but is further increased. When light travels from one medium to another medium, it is refracted and enters the new medium at a different angle. The degree of bending of the light's path depends on the angle that the incident beam of light makes with the surface of the prism, and on the ratio between the refractive indices of the two media

The index of refraction of a transparent material such as glass and water varies as a function of wavelength (**λ**). White light entering a prism is separated into different colors. This is how prisms and water droplets create rainbows from sunlight. This *spreading of light into its constituent spectral colors* is called (**dispersion)**.

A prism of glass separates white light into its spectral components in such a manner that colors associated with shorter wavelengths are more refracted than the colors

associated with longer wavelengths. So that “violet light” is less refracted than “red light”, as shown in figure (14)

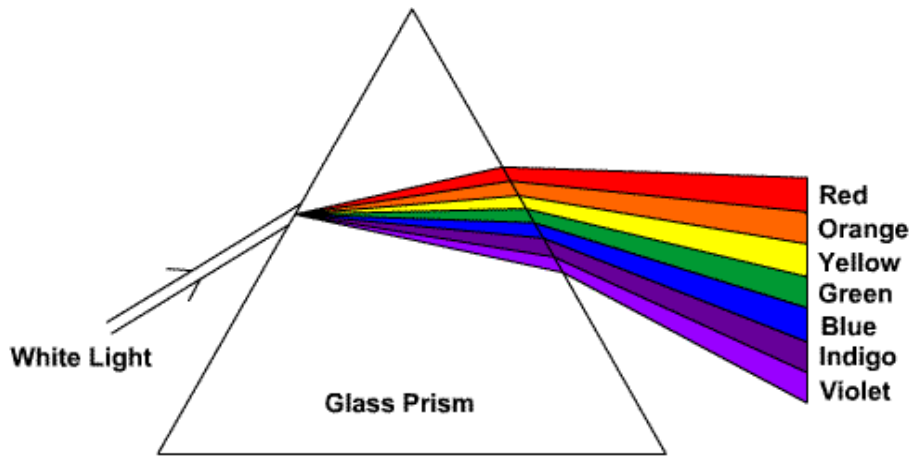


Figure (14): Color dispersion in prism

1-2-6-2 Derivation of the angle of Deviation (δ):

First let us consider, however, the geometrical optics of the prism for light of a single color, i.e., for *monochromatic light* such as is obtained from a sodium arc. According to the figure (15) we have:

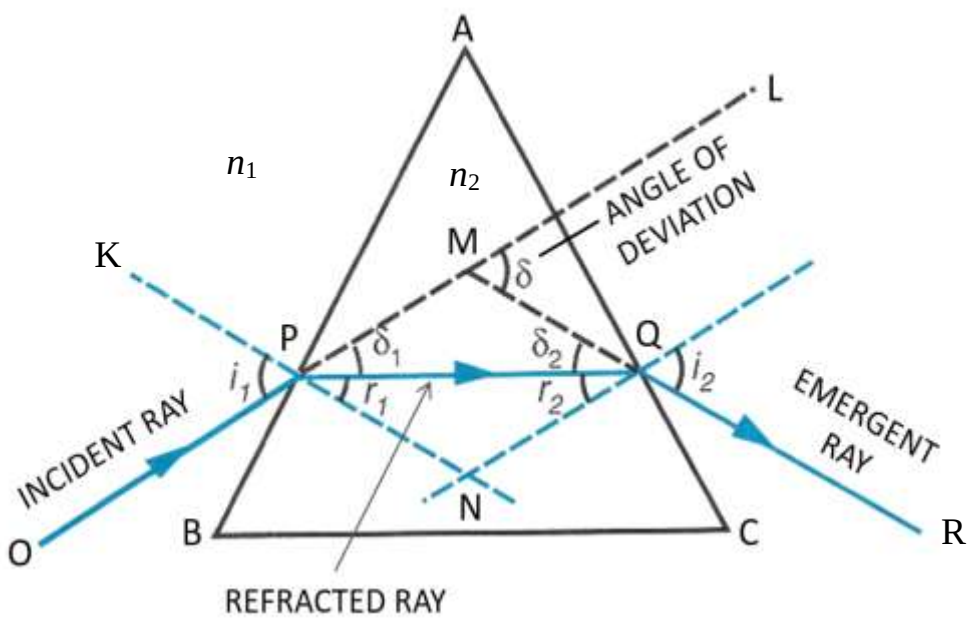


Figure (15): The geometrical optics of the prism

OP : is the incident ray on the prism.

PQ : is the refracted ray passing through the prism.

QR : the emergent ray coming out of the prism.

< A : is the prism angle.

< i_1 : is the incident angle.

< i_2 : is the emergent angle from the prism.

<r_1 : is the refracted angle from the first surface of the prism.

<r_2 : is the incident angle to the second surface of the prism.

δ = < LMR is the deviation angle.

δ_1 : is the deviation angle produced by the first surface AB

δ_2 : is the deviation angle produced by the second surface AC

Appling Snell's law for the face AB of the prism where refraction takes place from air to glass and the ray PQ will bend towards the normal KPN; we have:

$$\frac{n_2}{n_1} = \frac{sini_1}{sinr_1} \quad \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots \quad (30)$$

Appling Snell's law for the face AC of the prism where refraction takes place from glass to air and the emergent ray QR will bend away from the normal SQN; we have:

$$\frac{n_2}{n_1} = \frac{sini_2}{sinr_2} \quad \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots \quad (31)$$

In Δ MPQ;

< MPQ = $i_1 - r_1$ ……..…….(32)

Similarly :

< MQP = $i_2 - r_2$ ……..…….(33)

From exterior angle theorem on ΔMPQ we can write:

< LMR = δ = < MPQ + < MQP

$$= (i_1 - r_1) + (i_2 - r_2)$$

$$\delta = (i_1 + i_2) - (r_1 + r_2) \quad \ldots\ldots\ldots\ldots\ldots\ldots..(34)$$

From angle sum of Δ APQ :

$< A + < P + < Q = 180^o$ ……..…..(35)

And we have : $< APN = < AQN = 90^o$

We can write:

$< P = < APN - < QPN = (90^o - r_1)$ ……(36)

$< Q = < AQN - < PQN = (90^o - r_2)$ ……..(37)

Substituting eqs. (36) and (37) in eq. (35) we get:

$< A + (90^o - r_1) + (90^o - r_2) = 180^o$

$<A + 90^o - r_1 + 90^o - r_2 = 180^o$

$< A - r_1 - r_2 = 0$

$< A = r_1 + r_2$ ……..……(38)

Substituting eq. (38) and eq. (34) we have:

$\therefore \delta = (i_1 + i_2) - < A$

Or

$\boxed{\delta = (i_1 + i_2) - A}$ ..……(39)

So the angle of deviation of light ray in passing through a prism not only depend upon its material but also upon the angle of incidence.

2-2-6-2 Minimum angle of Deviation (δ_m):

If during the time a ray of light is refracted by a prism the prism is rotated continuously in one direction about an axis parallel to the refracting edge, the angle of deviation will be observed to decrease, reach a minimum, and then increase again, as shown in figure (16).

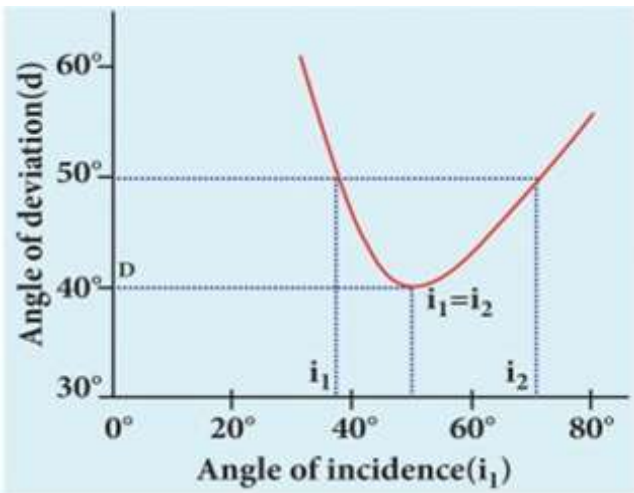


Figure (16): Plot between Angle of Deviation and Angle of incidence

The minimum value of angle of deviation is called *angle of minimum deviation* (δ_m). At minimum deviation:

(a) the angle of incidence is equal to the angle of emergence, ($i_1 = i_2$).

(b) the angle of refraction at the face one and face two are equal, ($r_1 = r_2$).

(c) the incident ray and emergent ray are symmetrical with respect to the prism.

(d) the refracted ray inside the prism is parallel to its base of the prism.

Refractive index of the material of the prism

At minimum deviation, $i_1=i_2=i$ and $r_1=r_2=r$

Equation (39) becomes,

$\delta = i_1 + i_2 - A \longrightarrow \delta = i + i - A \longrightarrow \boxed{\delta_m = 2i - A}$

$\therefore\ i = \frac{A+\delta_m}{2}$ ……..……..……(40)

The equation (38) becomes,

$r_1 + r_2 = A \longrightarrow r + r = A \longrightarrow 2r = A$

$\therefore \boxed{r = \frac{A}{2}}$ ……..(41)

Substituting i and r in Snell's law, where n_1=1, we have:

$$n_2 = \frac{sini}{sinr}$$

$$\boxed{n_2 = \frac{sin\left(\frac{A+\delta_m}{2}\right)}{\sin\left(\frac{A}{2}\right)}}$$ ……..(42)

The above equation is used to find the refractive index of the material of the prism. The angles A and δ_m can be measured experimentally.

For thin prism

In a thin or small angle prism, as the angles become very small, the sine of the angle nearly equals the angle itself.

As we mentioned earlier by applying Snell's law for the first face of the prism we have:

$\frac{n_2}{n_1} = \frac{sini_1}{sinr_1}$ becomes $n_2 = \frac{i_1}{r_1}$

$\therefore\ i_1 = n_2\, r_1$ ……(43)

And for the second face we have:

$\frac{n_2}{n_1} = \frac{sini_2}{sinr_2}$ becomes $n_2 = \frac{i_2}{r_2}$

$\therefore\ i_2 = n_2\, r_2$ ……..……(44)

Substituting eqs. (43) and (44) in eq. (39) we get:

$$\delta = i_1 + i_2 - A$$
$$= n_2\, r_1 + n_2\, r_2 - A$$
$$= n_2\,(r_1 + r_2) - A$$

$\because r_1 + r_2 = A$ (eq. (38))

Then we have:

$$\delta = n_2\,(A) - A$$

$$\boxed{\delta = A\,(n_2 - 1)}$$ ……..……..…..(45)

<u>Thus, it can be said that a thin prism is always in minimum deviation.</u>

Example (1): If the angle of minimum deviation of a light ray passing through a prism with prism angle 60° is 30°, then find the refractive index of the prism.

Solution:

We know that,

$$n_2 = \frac{\sin\left(\frac{A + \delta_m}{2}\right)}{\sin\left(\frac{A}{2}\right)}$$

We have, A=60° and δm = 30°

Substituting these values in the equation gives,

$$n_2 = \frac{\sin\left(\frac{60^o + 30^o}{2}\right)}{\sin\left(\frac{60^o}{2}\right)}$$

$$n_2 = \frac{\sin(45^o)}{\sin(30^o)} = 1.414$$

Example (2): The refractive index of the material of a prism is 1.5. When the prism is placed in a minimum deviation position, the angle of incidence is 45°. Calculate the angle of prism and the angle of minimum deviation.

Solution:

We have n_2=1.5, i =45° , A=? , δm =?

We know that for minimum deviation, $\delta m = 2i - A$

$\therefore 2i = A + \delta m$

$2\times45^o = A + \delta m$

$90^o = A + \delta m$

Also, we know that

$$n_2 = \frac{sin\left(\frac{A+\delta_m}{2}\right)}{\sin\left(\frac{A}{2}\right)} \qquad \text{then} \qquad \sin\left(\frac{A}{2}\right) = \frac{sin\left(\frac{A+\delta_m}{2}\right)}{n_2}$$

Substituting for 90° = A + δm we get,

$$\sin\left(\frac{A}{2}\right) = \frac{sin\left(\frac{90^o}{2}\right)}{1.5} = \frac{\sin 45^o}{1.5}$$

$$\frac{A}{2} = sin^{-1}(0.471) = 28.099^o$$

$$A \approx 2 \times 28^o$$

$$A = 56^o$$

Substituting this value of A in 90° = A + δm, we get,

$$\delta_m = 90^o - A$$

$$\delta_m = 90^o - 56^o$$

$$\delta_m = 34^o$$